DECOMPOSITION AND INERTIA GROUPS ARE SURJECTIVE ALONG THE RESTRICTION GAL(M/K) -> GAL(L/K)

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Allowed standard prerequisites (editorial).

- Every automorphism of the subextension L/K lifts to an automorphism of the Galois extension M/K. invoked at proof line 32890. Supplies tau in H mapping to the given sigma.
- The Galois group Gal(M/L) acts transitively on the primes of C lying over a fixed prime of B. invoked at proof lines 32892-32894. Provides sigma' correcting tau(r) to r.

Source reference. Original lemma 0BMK: sources/stacks-chow/tags.txt, source lines 14633–14633. Cited at proof line 32890 to lift sigma to tau in H; the conclusion is used inside the counted span.

Source reference. Original lemma 0BRK: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32777–32777. Cited at proof lines 32892-32894 to correct tau(r) back to r; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a normal domain with fraction field K. Let $M/L/K$ be a tower of (possibly infinite) Galois extensions of K. Let $H = Gal(M/K)$ and $G = Gal(L/K)$ and let C and B be the integral closure of A in M and L. Let $\mathfrak{r} \subset C$ and $\mathfrak{q} = B \cap \mathfrak{r}$. Set $D_\mathfrak{r} = \{\tau \in H \mid \tau(\mathfrak{r}) = \mathfrak{r}\}$ and $I_\mathfrak{r} = \{\tau \in D_\mathfrak{r} \mid \tau \bmod \mathfrak{r} = id_{\kappa(\mathfrak{r})}\}$ and similarly for $D_\mathfrak{q}$ and $I_\mathfrak{q}$. Under the map $H \to G$ the induced maps $D_\mathfrak{r} \to D_\mathfrak{q}$ and $I_\mathfrak{r} \to I_\mathfrak{q}$ are surjective.*

Proof. This is possible by Fields, Lemma 0BMK. Then $\tau(\mathfrak{r})$ and $\mathfrak{r}$ both lie over $\mathfrak{q}$. Hence by Lemma 0BRK there exists a $\sigma' \in \mathrm{Gal}(M/L)$ with $\sigma'(\tau(\mathfrak{r})) = \mathfrak{r}$. Hence $\sigma'\tau \in D_\mathfrak{r}$ maps to σ. The case of inertia groups is proved in exactly the same way using surjectivity onto automorphism groups. □

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